

BMIT2073 MOBILE APPLICATION DEVELOPMENT
WRITTEN TEST

Instructions: Answer all questions. Write your answers on separate sheets of paper.

Question 1

- a) (i) Construct a table to briefly compare between **native**, **hybrid** and **mobile web** apps based on the following **FIVE (5)** criteria:
- Development cost
 - App performance
 - Distribution method
 - Access to device features
 - Code maintenance
- (15 marks)
- (ii) Select the most appropriate mobile app type (**native**, **hybrid** or **mobile web**) for a highly interactive 3D mobile game. Briefly justify your selection. (2 marks)
- b) Identify the **FOUR (4) primary states** in Flutter app lifecycle (explanation is not required). In addition, provide a suitable task example for each of them. (8 marks)

[Total: 25 marks]

Question 2

- a) Provide **TWO (2)** differences between **stateless** and **stateful widgets**. (4 marks)
- b) Discover **THREE (3)** approaches for **efficient image loading** in relation to performance best practices for mobile apps. (6 marks)
- c) Present any **TWO (2)** methods of **internationalising** mobile apps for adapting to different languages, cultures and regions. Briefly elaborate your answers. (4 marks)
- d) Examine **TWO (2)** differences between **ephemeral** and **app states**. Then, provide an example for each of them. (6 marks)
- e) Identify the **FIVE (5) navigation principles** (explanation is not required). (5 marks)

[Total: 25 marks]